

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022-23

Course: B. Tech. Branch : Information Technology Semester : V

Subject Code & Name: BTITC502 [Computer Networks and Internetworking Protocols]

Max Marks: 60

Date: 31/01/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Explain the ISO OSI reference model.	CO1	6
B) i. An Ethernet MAC sublayer receives 1510 bytes of data from the LLC layer. Can the data be encapsulated in one frame? If not, how many frames need to be sent? What is the size of the data in each frame? ii. The data rate of 10Base5 is 10 Mbps. How long does it take to create the smallest frame? Show your calculation.	CO4	6
C) Compare and contrast CSMA/CD with CSMA/CA.	CO5	6
Q.2 Solve Any Two of the following.		12
A) Briefly define subnetting and supernetting. How do subnet mask and supernet mask differ from default mask in classful addressing?	CO3	6
B) Write short note on i. Repeater ii. Bridge iii. Router	CO4	6
C) i. An address in a block is given as 180.8.17.9. Find the number of addresses in the block, the first address, and the last address. ii. One of the addresses in a block is 167.199.170.82/27. Find the number of addresses in the network, the first address, and the last address.	CO3	6
Q. 3 Solve Any Two of the following.		12
A) Explain the Internet Control Message Protocol (ICMP) package in detail.	CO4	6
B) What is checksum? How checksum is calculated at sender and receiver?	CO5	6
C) Explain in detail ARP packet format.	CO1	6
Q.4 Solve Any Two of the following.		12
A) The following is a dump of a UDP header in hexadecimal format. CB84000D001C001C i. What is the source port number? ii. What is the destination port number? iii. What is the total length of the user datagram?	CO1	6

iv. What is the length of the data?		
v. Is the packet directed from a client to a server or vice versa?		
vi. What is the client process?		
B) Compare and contrast Transmission control protocol and User datagram Protocol	CO1	6
C) Explain transport layer services in detail.	CO1	6
Q. 5 Solve Any Two of the following.		12
A) The following is a dump of a TCP header in hexadecimal format. (05320017 00000001 00000000 500207FF 00000000) ₁₆	CO1	6
i. What is the source port number?		
ii. What is the destination port number?		
iii. What the sequence number?		
iv. What is the acknowledgment number?		
v. What is the length of the header?		
vi. What is the type of the segment?		
B) Explain TCP package in detail.	CO1	6
C) Explain the use of TCP timers in detail.	CO1	6

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech. Branch: Information Technology Semester: V
Subject Code & Name: BTITC502 [Computer Networks and Internetworking Protocols]
Max Marks: 60 Time: 02:00 PM To 05:00 PM Date:09/08/2023

Instructions to the Students:

1. All the questions are compulsory.
 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
 3. Use of non-programmable scientific calculators is allowed.
 4. Assume suitable data wherever necessary and mention it clearly.
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Q.1 Solve Any Two of the following. (12)

- | | |
|---|------------------------------|
| A) Define Internet Standards. Explain Maturity Levels of Request for Comment (RFC) in details. | Understand, (06)
Remember |
| B) Explain the OSI Model illustrating with suitable block diagram. | Analyze (06) |
| C) Describe Frame Format of an Ethernet along with Frame fields. | Understand, (06)
Apply |

Q.2 Solve Any Two of the following. (12)

- | | |
|--|--------------------------------------|
| A) An address in a block is given as 200.11.8.45. Find the number of addresses in the block, the first address, and the last address. | Create, (06)
Evaluate,
Analyze |
| B) Explain Loopback Addresses in details. | Apply (06) |
| C) Write Short Note on: | Understand, (06)
Remember |
| i) Network Mask | |
| ii) Subnetting | |
| iii) Address Aggregation | |

Q.3 Solve Any Two of the following. (12)

- | | |
|---|---------------------------|
| A) Explain Next-Hop method along with suitable block diagram. | Apply, (06)
Analyze |
| B) Describe Internet Protocol (IP) in details. | Analyze (06) |
| C) Define IP Package. Explain IP components with suitable block diagram. | Understand, (06)
Apply |

- Q.4 Solve Any Two of the following. (12)**
- | | | |
|---|-------------------------|------|
| A) Describe Hidden Station Problem with suitable example. | Analyze,
Evaluate | (06) |
| B) Explain Point to Point WANS in details. | Apply, Create | (06) |
| C) Explain Error Control mechanism within the Network Layer in brief. | Understand,
Evaluate | (06) |
- Q. 5 Solve Any Two of the following. (12)**
- | | | |
|---|------------------------------------|------|
| A) Change the following IPv4 addresses from binary notation to dotted-decimal notation.
a. 10000001 00001011 00001011 11101111
b. 11000001 10000011 00011011 11111111
c. 11100111 11011011 10001011 01101111
d. 11111001 10011011 11111011 00001111 | Analyze,
Evaluate | (06) |
| B) Find the error, if any, in the following IPv4 addresses:
a. 111.56.045.78
b. 221.34.7.8.20
c. 75.45.301.14
d. 11100010.23.14.67 | Understand,
Evaluate,
Create | (06) |
| C) Find the class of each address:
a. 227.12.14.87
b. 193.14.56.22
c. 14.23.120.8
d. 252.5.15.111 | Apply,
Analyze,
Evaluate | (06) |

***** ALL THE BEST *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/ Supplementary Winter Examination – 2023

Course: B. Tech. Branch: Information Technology Semester: V
Subject Code & Name: BTITC502 [Computer Networks and Internetworking Protocols]
Max Marks: 60 Date: 03 / 01 / 2023 Duration: 03 Hours

Instructions to the Students:

1. *All the questions are compulsory.*
 2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
 3. *Use of non-programmable scientific calculators is allowed.*
 4. *Assume suitable data wherever necessary and mention it clearly.*
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	(CO)	(Marks)
Q.1 Solve Any Two of the following.		(12)
A) Define Internet Standards. Explain Requirement Levels of Request for Comment (RFC) in details.	CO1, CO2	(06)
B) Explain the IEEE 802.11 services: the basic service set (BSS) and the extended service set (ESS) in details.	CO4	(06)
C) Describe Frame Format of an Ethernet along with Frame fields.	CO1, CO3	(06)
Q.2 Solve Any Two of the following.		(12)
A) Explain the Layers in the TCP/IP Protocol Suite illustrating with suitable block diagram.	CO2, CO6, CO4	(06)
B) Explain the types of Networks in Bluetooth Technology.	CO6, CO4	(06)
C) Write Short Note on:	CO1, CO2	(06)
i) Repeaters		
ii) Bridges		
iii) Routers		
Q.3 Solve Any Two of the following.		(12)
A) Define IP Package. Explain IP components with suitable block diagram.	CO3, CO4	(06)
B) An address in a block is given as 200.11.8.45. Find the number of addresses in the block, the first address, and the last address.	CO6, CO3, CO4	(06)
C) Explain Next-Hop method along with suitable block diagram.	CO1, CO3	(06)
Q.4 Solve Any Two of the following.		(12)
A) Explain the Occupation of the address space in classful addressing.	CO4, CO5	(06)

- | | | |
|---|----------|-------------|
| B) Describe Hidden Station Problem with suitable example. | CO3, CO6 | (06) |
| C) Explain the connection-oriented packet switched network illustrating with suitable block diagram. | CO1, CO5 | (06) |

Q. 5 Solve Any Two of the following. (12)

- | | | |
|--|---------------------|-------------|
| A) Change the following IPv4 addresses from binary notation to dotted-decimal notation.
a. 10000001 00011011 00011011 11101111
b. 11000001 10000011 00001011 11111111
c. 11100111 11011011 10001011 01101111
d. 11111001 10011011 11111011 00001111 | CO4, CO5 | (06) |
| B) Find the error, if any, in the following IPv4 addresses:
a. 331.56.45.78
b. 172.34.07.8.20
c. 75.45.300.14
d. 11101010.23.14.67 | CO1,
CO5,
CO6 | (06) |
| C) Find the class of each address:
a. 227.12.14.87
b. 11.14.56.22
c. 172.23.120.8
d. 251.5.15.111 | CO3,
CO4,
CO5 | (06) |

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**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE**

Supplementary Summer Examination – 2024

Course: B. Tech. Branch: Information Technology Semester: V

Subject Code & Name: BTITC502_Y22

Subject Name: Computer Networks and Internetworking Protocols

Max Marks: 60

Date: 03/07/2024

Duration: 3 Hrs

Instructions to the Students:

1. *All the questions are compulsory.*
 2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
 3. *Use of non-programmable scientific calculators is allowed.*
 4. *Assume suitable data wherever necessary and mention it clearly.*
-

	(CO)	(Marks)
Q.1 Solve Any Two of the following.		(12)
A) Define Internet Standards. Explain types of Internet service providers (ISPs).	CO1, CO2	(06)
B) Explain the IEEE 802.11 services: the basic service set (BSS) and the extended service set (ESS) in details.	CO4	(06)
C) Describe the Maturity Levels of an RFC.	CO1, CO3	(06)
Q.2 Solve Any Two of the following.		(12)
A) Explain the Layers in the TCP/IP Protocol Suite illustrating with suitable block diagram.	CO2, CO6, CO4	(06)
B) Explain the types of Networks in Bluetooth Technology.	CO6, CO4	(06)
C) Write Short Note on:	CO1,	(06)
i) Repeaters	CO2	
ii) Bridges		
iii) Routers		
Q.3 Solve Any Two of the following.		(12)
A) Explain the Layered Architecture of OSI Layers with suitable block	CO3, CO4	(06)

diagram.

- | | | |
|--|------------------|-------------|
| B) An address in a block is given as 200.11.8.45. Find the number of addresses in the block, the first address, and the last address. | CO6, CO3,
CO4 | (06) |
| C) Explain Next-Hop method along with suitable block diagram. | CO1, CO3 | (06) |

Q.4 Solve Any Two of the following. (12)

- | | | |
|---|----------|-------------|
| A) Describe Hidden Station Problem with suitable example. | CO4, CO5 | (06) |
| B) Explain the Occupation of the address space in classful addressing. | CO3, CO6 | (06) |
| C) Explain the connection-oriented packet switched network illustrating with suitable block diagram. | CO1, CO5 | (06) |

Q.5 Solve Any Two of the following. (12)

- | | | |
|--|----------|-------------|
| A) Change the following IPv4 addresses from binary notation to dotted-decimal notation. | CO4, CO5 | (06) |
|--|----------|-------------|

a. 10000001 00011011 00011011 11101111

b. 11000001 10000011 00001011 11111111

c. 11100111 11011011 10001011 01101111

d. 11111001 10011011 11111011 00001111

- | | | |
|--|------|-------------|
| B) Find the error, if any, in the following IPv4 addresses: | CO1, | (06) |
|--|------|-------------|

a. 331.65.54.87

CO5,

b. 172.43.07.7.13

CO6

c. 75.45.400.14

d. 11101010.23.14.67

- | | | |
|---|------|-------------|
| C) Find the class of each address: | CO3, | (06) |
|---|------|-------------|

a. 227.12.14.87

CO4,

b. 11.14.56.22

CO5

c. 172.23.120.8

d. 251.5.15.111

***** ALL THE BEST *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech.

Branch : Information Technology

Semester : V

Subject Code & Name: BTITC502, Computer Networks and Internetworking Protocols

Max Marks: 60

Date: 08/02/2025

Duration: 3 Hrs.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 is compulsory and include objective-type questions.
3. Students are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level /CO)	Marks
Q. 1 Objective type questions (Compulsory Question).		12
1 ----- is responsible for source to destination delivery of a packet.	(CO4)	1
a. Data Link Layer b. Network Layer c. Transport Layer d. Application Layer		
2 What is the transmission speed of the Standard Ethernet?	(CO4)	1
a. 10 Mbps b. 100 Mbps c. 1 Gbps d. 10 Gbps		
3 Which address is used by a router to send a packet to all hosts in a specific network?	(CO5)	1
a. Network Broadcast b. Limited Broadcast c. All Zeros d. Direct Broadcast		
4 In supernetting, an organization can combine several ----- blocks to create a larger range of addresses.	(CO3)	1
a. Class A b. Class B c. Class C d. Class D		
5 Class D address is used for -----	(CO3)	1
a. Broadcast b. Limited Broadcast c. Unicast d. Multicast		
6 After the server TCP receives a passive open request from the server application program, it goes to the ----- state.	(CO1)	1
a. CLOSED b. LISTEN c. SYN-RCVD d. ESTABLISHED		
7 Which of these is not a type of error-reporting message?	(CO5)	1
a. Destination unreachable b. Source quench c. Router advertisement d. Time exceeded		
8 What statement is true regarding ICMP packets?	(CO5)	1
a. They are encapsulated within UDP datagram b. They acknowledge receipt of TCP segments c. They guarantee datagram delivery d. They are encapsulated within IP datagrams		
9 The input and output ports of a router perform the ----- layer functions of the router.	(CO4)	1
a. physical and data link b. network c. transport d. none of the these		

10	The ----- defines the client program.	(CO2)	1
	a. ephemeral port number b. IP address c. well-known port number d. physical address		
11	Which component in UDP communicates with ICMP software package?	(CO1)	1
	a. control block b. control module c. input module d. output module		
12	----- timer is needed to handle the zero window size advertisement.	(CO1)	1
	a. retransmission b. persistence c. keep alive d. time-waited		
Q.2 Solve the following.			
A)	Explain in brief the protocols at internetworking layer in the TCP/IP protocol suite.	(CO1)	6
B)	What is Gigabit Ethernet? Show its implementations.	(CO6)	6
Q.3 Solve the following.			
A)	Describe direct broadcast and loopback special addresses with example networks.	(CO3)	6
B)	Explain how to find the network address, the number of addresses and the last address in the block of classless addressing, if one of the addresses is given with CIDR notation. Hence, find the above information if one of the addresses in a block is 167.199.170.82/27.	(CO4)	6
Q.4 Solve Any Two of the following.			
A)	Explain the fields which are related to fragmentation and reassembly of an IP datagram.	(CO4)	6
B)	Explain structure of a router with its components. How the problem of internal collision with the Banyan switch is solved using Batcher-Banyan switch?	(CO5)	6
C)	Draw simplified ICMP package diagram, showing its components and their interactions. Write output module.	(CO5)	6
Q.5 Solve Any Two of the following.			
A)	Write two solutions to prevent the silly window syndrome created by the receiver.	(CO1)	6
B)	Describe checksum calculation in UDP. How it differs from IP?	(CO1)	6
C)	Comment on congestion policy of TCP and the relationships between its phases.	(CO5)	6

Q. 6 Solve Any Two of the following.

12

- A) Write the most common rules for generating acknowledgements in TCP. (CO2) 6
- B) Explain with single time-line diagram the scenario of connection establishment, data transfer and half-close termination in TCP. (CO1) 6
- C) What are the features of UDP? (CO1) 6
Write some typical applications that can benefit more from the services of UDP than TCP.

End